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ODD END SEMESTER EXAMINATION DECEMBER - 2024

Name of the Course: B. Tech

Semester: VII

Name of the Paper: Cloud Orchestration and Load Balancing Paper Code: TCS 750

Time: 3 Hours

Maximum Marks: 100

Note: -

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	CO1
(a)	Derive Kubernetes facilitation for orchestration in modern cloud environments with suitable example.	
(b)	Elaborate the effective resource scheduling contribution for cost optimization in cloud computing with suitable example.	
(c)	Assess automation in cloud computing along with atleast five benefits that it offers to organizations.	
Q2	(20 marks)	CO2
(a)	How can middleware be used to integrate different cloud services and ensure smooth communication between them? Explain in detail.	
(b)	A company is moving its infrastructure to the cloud and needs to manage a wide range of resources, including storage, computing power, and network management. Explain how resource provisioning can be automated to scale up or down based on the company's needs?	
(c)	Discuss how middleware, resource provisioning, metering, and orchestration collectively ensure efficient cloud management. Use a real-world example to support your explanation.	
Q3	(20 marks)	CO3, CO6
(a)	Compare the considerations for cloud deployment on-premises versus in a public cloud. How does the choice of deployment impact scalability and fault tolerance?	
(b)	Using Microsoft Azure Automation or Morpheus, describe a scenario where the tool can enhance scaling strategies, optimize cost, and improve fault tolerance.	
(c)	Evaluate how the orchestrator handles scaling, load balancing, and fault tolerance in cloud deployments.	
Q4	(20 marks)	CO4
(a)	Investigate the Heat Orchestration Service in OpenStack. Describe its primary functions in cloud resource management.	
(b)	Outline the main execution flow in the MapReduce model. What scheduling techniques are commonly used in MapReduce?	
(c)	MapReduce, Spark, and GraphLab are popular cloud programming models. Analyze a data-intensive scenario (log analysis or recommendation system) and justify the most suitable model to use, considering execution flow, scheduling, and fault tolerance.	
Q5	(20 marks)	CO5
(a)	Your REST interfaced web service has been deployed on the cloud and is being tested using a load generator for a fixed time. Explain the results you would expect in terms of throughput, latency, and resource usage.	
(b)	Investigate tools, frameworks, or libraries will be used to implement the web-service and run the analytics jobs? Provide justifications for your choices.	
(c)	Propose a method to measure and log the cloud resources used during the load test (e.g., CPU, memory, network). Justify your choice also.	

